Exam 01

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1 Student Information & Instructions

Student ID number: Family Name: Prename:

EXAM: Introduction to Analysis February 2nd, 2026

- The use of electronic devices or documents is not permitted.
- Put your answers onto these sheets.
- Keep these sheets stapled together.
- Any additional sheets will not be graded.
- Multiple-Choice-Part:

Each multiple-choice question has exactly one correct answer. Points are awarded only if the correct answer is marked; no points are given for incorrect or missing answers. No points are deducted for wrong answers.

2 Problem 1

(3 points) Define $d(x,y) := |\frac{x}{1+|x|} - \frac{y}{1+|y|}|$ $\quad d_0(x,y) = \begin{cases} 1, & x \neq y, \\ 0, & x = y, \end{cases}, \quad x, y \in \mathbb{R}$

(a) Prove that d is a metric on $\mathbb{R}$.

(b) Provide an explicit sequence $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ that converges with respect to d but does not converge in the metric do.

(a) Let x, y, $z \in \mathbb{R}$ be arbitrary. Then, $d(x,y) \geq 0$ and symmetry holds by $d(x,y) = |\frac{x}{1+|x|} - \frac{y}{1+|y|}| = |(-1)(\frac{x}{1+|x|} + \frac{y}{1+|y|})| = |-\frac{x}{1+|x|} + \frac{y}{1+|y|}| = d(y,x)$

Triangle inequality: $d(x,y) = |\frac{x}{1+|x|} - \frac{z}{1+|z|} + \frac{z}{1+|z|} - \frac{y}{1+|y|}| \leq |\frac{x}{1+|x|} - \frac{z}{1+|z|}| + |\frac{z}{1+|z|} - \frac{y}{1+|y|}| = d(x,z) + d(z,y)$.

If $x = y$, then $d(x,y) = d(x,x) = 0$. Definite: If $d(x,y) = 0$, then $\frac{x}{1+|x|} = \frac{y}{1+|y|}$ This implies $x(1+|y|)=y(1+|x|).$

Case x, $y \geq 0$: Then $x + xy = y + yx$ and hence $x=y.$ Case x, $y \leq 0$: Then $x - xy = y - yx$ and hence $x = y$ Case $x > 0$, $y < 0$ Then $x - xy = y + yx$ and hence $0 < x = y + 2xy < 0$ contradiction. Case $x < 0$, $y > 0$: leads also to a contradiction by symmetry.

(b) Define $x_n = \frac{1}{n}$ $n \in \mathbb{N}\backslash\{0\}$. Then $d(x_n, 0) = \frac{1/n}{1+1/n} = \frac{1}{n+1} \to 0$, which implies convergence towards 0 with respect to the metric d. We also observe $d_0(x_n, 0) = 1$ and there cannot exist any $x \in \mathbb{R}$ such that $d_0(x_n, x) \to 0$.

3 Problem 2

(3 points) Consider the function $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = \log x - x^2$.

(a) Compute all local extrema. (b) Decide and reason whether f has a global maximum on (0,0).

(a) $f'(x) = \frac{1}{x} - 2x = 0$ implies $\frac{1}{x} = 2x$ and hence $1 = 2x^2$. We have $x_0 = 1/\sqrt{2}$ and the other solution is not in $(0, \infty)$. $f''(x_0) = -\frac{1}{x_0^2} - 2 = -2 - 2 = -4 < 0$ implies that x_0 is a local maximum.

(b) f' is positive in $(0, x_0)$ and negativ in (x_0, ∞). Thus, f is increasing in $(0, x_0)$ and decreasing in (x_0, ∞). Therefore, the local maximum is a global one.

4 Problem 3

(3 points) Consider the functions $f(x) = x^x$ for $x > 0$ Determine the Taylor polynomial of f of degree 1 centered at $x_0 = 1$.

$f(x) = x^x = \exp(x \log x)$ so that

$$f'(x) = \exp(x \log x)(1 \log x + x\frac{1}{x}) = x^x(\log x + 1)$$

Hence, $f(1) = 1 = f'(1)$ and the Taylor polynomial of degree 1 in $x_0 = 1$ is

$$T_{f,1,1}(x) = f(x_0) + f'(x_0)(x - x_0) = 1 + (x - 1) = x$$

5 Problem 4

(3 points) Let $f : [0,1] \to \mathbb{R}$ be given by $f(x) = 84x$.

(a) Compute the lower Darboux-sum with respect to the equidistant partition $\mathcal{P}_n$ for $n = 1,2,\ldots$.
(b) Compute the upper Darboux-sum with respect to the equidistant partition $\mathcal{P}_n$ for $n = 1,2,\ldots$.
(c) Determine the integral value by taking limits of the Darboux sums.

The equidistant partition is $\mathcal{P}_n = \{\frac{k}{n} : k = 0, 1, ..., n\}$. The function f is increasing.

(a) The lower Darboux sum is

$$L(f, \mathcal{P}_n) = \frac{1}{n}\sum_{k=1}^{n} f(\frac{k-1}{n}) = \frac{84}{n}\sum_{k=1}^{n}\frac{k-1}{n} = \frac{84}{n^2}\sum_{k=1}^{n}(k-1) = \frac{84}{n^2}\frac{(n-1)n}{2} = 42\frac{n-1}{n}$$

(b) The upper Darboux sum is

$$U(f, \mathcal{P}_n) = \frac{1}{n}\sum_{k=1}^{n} f(\frac{k}{n}) = \frac{84}{n}\sum_{k=1}^{n}\frac{k}{n} = \frac{84}{n^2}\sum_{k=1}^{n}k = \frac{84}{n^2}\frac{n(n+1)}{2} = 42\frac{n+1}{n}$$

(c) Since $L(f, \mathcal{P}_n) = 42\frac{n-1}{n} \to 42$ and $U(f, \mathcal{P}_n) = 42\frac{n+1}{n} \to 42$, we have

$$\int_0^1 f(x)dx = 42$$

6 Problem 5

(3 points) Compute the following integrals: (a) $\int_0^1 x\ e^{(x^2)} dx$ (b) $\int_0^1 x\ e^{-x} dx$

(a) The substitution $u = x^2$ yields $\frac{du}{dx} = 2x$ According to $dx = \frac{du}{2x}$ we obtain

$$\int_0^1 x\ e^{(x^2)} dx = \frac{1}{2} \int_0^1 e^u du = \frac{1}{2}[e^u]_0^1 = \frac{1}{2}(e^1 - e^0) = \frac{1}{2}(e - 1)$$

(b) Partial integration leads to

$$\begin{aligned}
\int_0^1 x\ e^{-x} dx &= [-xe^{-x}]_0^1 - \int_0^1 -e^{-x} dx \\
&= -e^{-1} + 0e^0 + [-e^{-x}]_0^1 \\
&= -1/e - e^{-1} + e^0 \\
&= 1 - 2/e
\end{aligned}$$

7 Problem 6

(3 points) Let $g : \mathbb{R} \to \mathbb{R}$ be functions that are continuous at a point $x_0 \in \mathbb{R}$ Assume that f is differentiable at x_0 and that $f(x_0) = 0$.

Prove that the product $f \cdot g$ is differentiable in x_0.

Since g is continuous in x_0 and f is differentiable in x_0, we observe

$$\begin{aligned}
\lim_{x \to x_0} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} &= \lim_{x \to x_0} \frac{f(x)g(x)}{x - x_0} \\
&= \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} g(x) \\
&= \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \lim_{x \to x_0} g(x) \\
&= f'(x_0)g(x_0)
\end{aligned}$$

Hence, the product $f \cdot g$ is differentiable in x_0.

8 Problem 7

Multiple-Choice-Reminder: There is only one correct answer per question.

(1 point) The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is

- ☐ divergent
- ☐ absolutely convergent
- ☒ convergent
- ☐ convergent towards 0

9 Problem 8

(1 point) Which of the following functions is convex on R?

- ☐ $f(x) = \sin x$
- ☐ $f(x) = x^3$
- ☒ $f(x) = x^4 + x^2$
- ☐ $f(x) = \cos x$

10 Problem 9

(1 point) Given a sequence of differentiable functions $f_n : [a, b] \to \mathbb{R}$ If $(f_n)_{n\in\mathbb{N}}$ converges towards $f : [a, b] \to \mathbb{R}$ in the sup sup-norm $d_\infty(f_n, f) = \sup_{x\in[a,b]} |f_n(x) - f(x)| \to 0$ what is necessarily true:

- ☐ f is constant
- ☒ f is continuous
- ☐ f is differentiable
- ☐ f has at least one local extrema

11 Problem 10

(1 point) Which rule does this Julia code implement in order to approximate the integral?

```
function myJuliaFunction(f,a,b; n=100)
x= range (a,b,n+1)
h=(b-a)/n
y=x.+h/2
return sum(f(y[k]) for k=1:n)*h
end
```

- ☐ trapezoidal rule
- ☒ mid-point rule
- ☐ lower Darboux sum
- ☐ upper Darboux sum